

1. Executive Summary

As part of our corporate finance group assignment, we conducted a comprehensive investment appraisal and valuation of PJSC Polyus, the largest gold producer in Russia and a global top-5 player. We focused on evaluating the "Sukhoi Log" deposit - one of the largest undeveloped gold deposits globally. To capture the long-term value of this standalone asset, we constructed a 25-year financial model (2025–2050), testing the project's viability through Free Cash Flow to the Firm (FCFF), Free Cash Flow to Equity (FCFE), and Adjusted Present Value (APV) approaches. Our analysis indicates a highly robust project with an IRR of 22.38% and a Base NPV of \$32.08 billion, leading to a confidently bullish outlook for potential investors.

2. Industry & Macroeconomic Environment

The project is being implemented in a highly complex macroeconomic environment characterized by elevated geopolitical (country) risks and shifting supply chains. However, Polyus operations are naturally hedged by the underlying asset: gold.

Historically, gold acts as a safe-haven asset. To project future revenues, we deliberately avoided short-term market euphoria. Instead, we analyzed gold prices over a 50-year horizon (the post-Bretton Woods era, 1975–2025). This analysis yielded a historical Compound Annual Growth Rate (CAGR) of 8.64%. We applied this steady growth rate starting from the average gold price of 2025 (3,431/oz).

As a cross-check, our 2026–2027 generated prices (3,727 and \$4,049/oz) are significantly lower than current optimistic forecasts by major investment banks like J.P. Morgan, proving that our revenue modeling is highly conservative, providing a built-in margin of safety.

3. Project Overview and Key Assumptions

Sukhoi Log is currently in its capital-intensive development phase. Based on company guidance and our internal assumptions, our model is built on the following parameters:

- **Production Volume:** We assumed a flat, conservative recovery/production rate of 2.014 million ounces per year beginning in 2030, which is below the upper bound of the company's 2.3 Moz/year guidance.
- **Operating Costs:** Polyus maintains absolute cost leadership. We modeled the Total Cash Cost (TCC) starting at \$653/oz in 2025. SG&A and Other Expenses were fixed conservatively at 6% and 5.2% of revenue, respectively.
- **Capital Expenditures (CapEx):** The project is front-loaded with massive investments. We modeled 2.3 billion in 2025 and 1.5 billion annually from 2026

to 2029, funded entirely through project finance (debt) to maximize the leverage effect.

- Taxation: A critical driver of value in our model is the Russian Regional Investment Projects (RIP) framework. We modeled preferential tax treatment: Profit tax remains at 0% in the early production years, stepping up to 10% (2034-2038), and finally normalizing at 25% from 2039 onwards. We also incorporated the Mineral Extraction Tax (MET), including current surcharges.

4. Cost of Capital & Beta Evaluation

A major challenge for our group was determining the appropriate discount rate. We tested several methodologies to estimate the equity beta for Polyus:

1. Standard Beta (IMOEX): Yielded an equity beta of 0.78.
2. Standard Beta (GDM Index / Gold Futures): Yielded betas of 0.40 and 0.57, respectively.
3. Industry Beta from Developed Markets: By analyzing peers (Barrick Gold, Newmont, Agnico Eagle), we found an average industry unlevered beta of ~0.677.

Our Choice & Justification: We concluded that using standard regression against the local Russian market overestimates systemic risk due to isolation, while ignoring the defensive nature of gold. To accurately assess the required return, we employed Aswath Damodaran's framework.

Using an unlevered beta adjusted for local leverage (yielding a levered beta of 0.40), a base Risk-Free rate of 13.00%, an Equity Risk Premium (ERP) of 4.41%, and a critical Country Risk Premium (CRP) of 3.90%, we calculated a Cost of Equity (R_e) of 18.66%. Combining this with a 10% Cost of Debt, our resulting WACC fluctuates insignificantly around ~17.2% – 17.4% across the project life. This exceptionally high discount rate fairly penalizes the project for current geopolitical and operational risks in Russia.

5. Valuation Approaches & Results

To provide an objective valuation, our group modeled the cash flows using three distinct methodologies. Despite the aggressive 17.4% discount rate, the project generates tremendous fundamental value:

- WACC Approach (Base NPV): \$32.08 billion. By discounting the Free Cash Flow to the Firm (FCFF) at the WACC, we captured the total enterprise value added by the new mine.
- Adjusted Present Value (APV): 30.85 billion. We consider APV the most mathematically elegant method for a project of this scale. By discounting unlevered cash flows at the Unlevered Cost of Equity ($V_u = 30.29B$) and

adding the Present Value of Tax Shields (PVTS = 556million) generated by the 8.3 billion in cumulative debt, we precisely isolated the value created by financial engineering.

- Flow to Equity (FTE): \$29.36 billion. Discounting FCFE at the 18.66% Cost of Equity yields a slightly lower, but highly correlated, shareholder value.

Additional Metrics: The project boasts a Profitability Index (PI) of 4.87 and a Discounted Payback Period of just 6.19 years. Given that operations only begin in Year 5 (2030), this means the project recovers all its multi-billion-dollar discounted investments almost immediately after the factory reaches full capacity.

6. Investor Conclusion

From a potential investor's point of view, PJSC Polyus and its Sukhoi Log project present a compelling "High Potential / High Risk" opportunity.

While the upfront capital requirements are immense and lack of gold hedging leaves the company fully exposed to commodity cycles, our highly conservative pricing model proves that the asset's low-cost nature provides an immense margin of safety. The project's stellar IRR of 22.38% comfortably clears the high 17.4% hurdle rate (WACC).

Recommendation: For investors with an appropriate risk appetite and a long-term horizon, the successful realization of Sukhoi Log will profoundly transform Polyus's cash-generation capabilities. We initiate a BUY recommendation, viewing the company's equity as highly undervalued relative to the cash flows this uncompleted project will begin generating in the 2030s.